

Akshay Varma

Gen AI Engineer

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PROFILE

Generative AI engineer with 4+ years of experience, including more than 2 years building and scaling large language model products across retrieval augmented generation, agentic workflows, evaluation, and safety. Owns the full lifecycle from discovery to proof of concept to general availability to monetization, partnering with product, security, platform, and design. Hands on with python, PyTorch and Transformers, LangChain and LangGraph, vector databases, Docker, CI/CD, and AWS. Delivers state-of-the-art and secure systems with strong governance while mentoring engineers and influencing stakeholders.

CORE COMPETENCIES

- **Programming Languages:** Python, R, SQL
- **Data Science:** Machine learning, Deep learning, Natural Language Processing (NLP), PyTorch, Tensorflow, Bedrock
- **GenAI & LLMs:** Fine tuning, Retrieval Augmented Generation (RAG), LangChain, LangGraph, Agentic AI, Chatbots, AI assistants, Prompt engineering, Transformers, Hugging face
- **Retrieval & Graph:** Vector databases, FAISS, Pinecone, OpenSearch, embeddings BGE and E5
- **Evaluation & Safety:** Golden sets, Rubric scoring, Precision, recall & ROUGE, A/B testing, Jailbreak prompting, PII guardrails, Vulnerability analysis
- **Data Engineering:** SQL, NoSQL, Apache Spark, dbt, Kafka, Snowflake
- **MLOps/LLMOps/DevOps:** Docker, GitHub, GitLab, CI/CD pipelines, LangSmith, Splunk, vLLM
- **Cloud Platforms:** AWS (Bedrock, SageMaker, Redshift, S3, IAM, ECS and ECR), GCP
- **Soft Skills:** Leadership, Attention to detail, Continuous learning, Storytelling, Curiosity, Team collaboration

PROFESSIONAL EXPERIENCE

Senior GenAI Engineer

Workhuman

10/2024 – Present

Dublin, Ireland

Workhuman builds human recognition and culture products. Led the transformation of recognition data into Human Intelligence through an AI assistant focused on skills, awards, and hidden talent discovery.

- Developed & launched the Recognition Intelligence Assistant from PoC to GA and monetized it; converted **8%** of existing clients to the premium tier and **secured 3 new enterprise contracts** for org-wide rollout.
- Introduced chain-of-thought and few-shot prompting with self-consistency and token budgeting; raised exact match accuracy **14%** and reduced tokens per response by **about 200 tokens**.
- Built RAG on recognition and policy content with **hybrid retrieval BM25 plus BGE embeddings, FAISS ANN, and bge cross-encoder re-ranking**; improved factual hit rate **18 to 24 percent** and cut **p95** latency from **~850 ms to ~560 ms**.
- Engineered **Bedrock** backed services with Anthropic **Claude 3.7 Sonnet** behind **FastAPI** and **Docker** with **Splunk** observability; sustained throughput over **90,000 tokens per minute**.
- Built **LangChain** and **LangGraph** based agent flows and a **history-aware multi-turn chat**; cut prompt rollbacks from 5 to 2 per quarter and reduced average turns to resolution from about 3 to 2.
- Implemented **guardrails** and governance including PII redaction, policy-based refusals, and jailbreak tests on Redshift and S3 backed evaluation datasets; achieved **zero P0 security incidents**, reduced A/B cycle time from 9 days to 5, and **lowered production data defects** by over 40%.
- Mentored a pod of three to five engineers, authored confluence pages for prompt versioning and rollback and retrieval quality bars.

Data Scientist

Rita Data

07/2022 – 09/2024

Amsterdam, Netherlands

Rita is a zero-party data platform that turns user shared data into privacy safe insights for brands. Shipped consent aware assistants and retrieval systems with strict governance.

- Developed and launched an AI-powered SaaS **market analytics platform** with a **zero-party data marketplace**; moved from first prototype to three paying pilots in **120 days** and cut customer activation time from **2 weeks** to **4 days**.
- Implemented a consent driven RAG pipeline over partner policies and user data using OpenSearch with FAISS and Pinecone, plus cross encoder re-ranking and HNSW tuning to lift relevant hit rate and increase query throughput.
- Shipped a **GPT-powered** competitive market insights platform with **LangChain** and **Snowflake** that enabled self-serve answers and reduced analyst tickets by roughly 33%.
- Built **NLP text classification** for topic, intent, sentiment, and claim extraction on **PyTorch Transformers**; achieved **F1 above 95** on internal sets and batch scored **500k documents per day** on **AWS**.
- Deployed client-facing conversion and churn models; achieved **AUC 0.86**, lifted trial-to-paid conversion **by ~7%**, and cut churn rate **20%**.

Software Development Engineer

Accenture

12/2019 – 12/2020

Mumbai, India

- Developed Salesforce features with automated tests and continuous delivery. Reduced post release defects and trimmed cycle time in successive sprints.
- Standardized logging and deployment runbooks across teams to stabilize releases.

PROJECT PORTFOLIO

Human Intelligence AI Assistant (*Workhuman*)

- **Description:** HR copilot surfacing skills, drafting recognitions, and recommending mentors and awards with retrieval augmented generation and agentic flows.
- **Impact:** Premium conversion of existing client's 8%; factual hit rate plus 14%; p95 latency about 850 milliseconds to around 560 milliseconds; throughput over 90,000 tokens per minute.
- **Stack:** Claude Sonnet 3.7 on Bedrock, LangChain and LangGraph, FAISS and bge reranker, FastAPI, Docker, Splunk.
- **Role:** Prompt engineering, Software development, RAG architecture, guardrails, and offline and online evaluation.

Manager 1:1 Talent Briefs Copilot (*Workhuman*)

- **Description:** History aware assistant compiling weekly briefs from recognition, goals, etc. with semantic memory.
- **Impact:** Turns to resolution down about one third, nomination cycle shorter by two business days.
- **Stack:** LangGraph memory, policy filters, semantic session store, Redshift and S3 evaluation tables.
- **Role:** Owner for orchestration, prompt strategy, guardrails, and retrieval quality; defined golden sets and continuous integration checks.

AI-Powered Market Analytics Platform with Zero-party Data Marketplace (*Rita Data*)

- **Description:** B2B SaaS that combines a zero-party data marketplace with consent-aware RAG and a natural-language analytics assistant that generates SQL on Snowflake for market and competitor insights.
- **Impact:** From prototype to 3 paying pilots in 120 days; reduced activation time from 2 weeks to 3 days; manual analysis requests down by ~33% saving considerable hours per quarter.
- **Stack:** Snowflake with dbt, DynamoDB, S3, Kafka; LangChain, Pinecone, OpenSearch; OpenAI GPT; Streamlit.
- **Role:** Lead for data engineering and science (pipelines, quality, ingestion, models, governance, etc.); client onboarding, owner for prompts, quality rubric, and release.

EDUCATION

B.E. Electronics & Telecommunications Engineering – Savitribai Phule Pune University, India

Graduated: 06/2019